

Module 2: Frequency, Probability, and Distributions — Summary

What this module is about. A statistical study begins with a group we want to understand, but we usually observe only part of that group. Counts from the observed sample can be converted into relative frequencies, which can be interpreted as empirical probabilities. The way data are displayed depends on whether the variable consists of named categories or numerical values. A probability distribution must assign nonnegative probabilities that add to 1.

- The **population** is the entire group the question is about.
- The **sample** is the part of the population actually observed.
- Every realized sample is only a partial view of the population and will differ from it in some ways.
- The practical question in this course is whether the **selection method systematically overrepresents or underrepresents** parts of the population.

Random selection is useful because it gives a broader cross-section of the population than selecting people only from one convenient time or place.

Example 1: From a frequency table to probabilities

Twenty employees report how they commute to work: 10 drive, 4 take the bus, 3 bike, and 3 walk. The total sample size is $n = 20$.

$$P(\text{drive}) \approx \frac{10}{20} = 0.50, \quad P(\text{bus}) \approx \frac{4}{20} = 0.20.$$

The four relative frequencies add to 1:

$$0.50 + 0.20 + 0.15 + 0.15 = 1.00.$$

Because commuting method is categorical, a bar chart is a natural display.

If category i occurs f_i times in a sample of size n , its relative frequency is

$$r_i = \frac{f_i}{n}.$$

A complete probability distribution satisfies

$$p_i \geq 0 \quad \text{for every outcome,} \quad \sum p_i = 1.$$

The complement rule is

$$P(\text{not } A) = 1 - P(A).$$

- Use a **bar chart** for named categories such as car, bus, bike, and walk. Bars are separated because the categories are distinct names.
- Use a **histogram** for quantitative data grouped into numerical intervals. Bars touch because the intervals form a numerical scale.

Example 2: Repairing a probability distribution

Suppose four outcomes have probabilities 0.15, 0.30, 0.25, and p . Because the total must be 1,

$$0.15 + 0.30 + 0.25 + p = 1.$$

The known probabilities sum to 0.70, so $p = 0.30$. If event A has probability 0.30, then

$$P(\text{not } A) = 1 - 0.30 = 0.70.$$

By the end of this module, you should be able to: identify a population and sample; describe how a sampling method may overrepresent or underrepresent part of a population; build a frequency and relative-frequency table; choose a bar chart or histogram appropriately; verify a probability distribution; and use complements.

Module 2: Frequency, Probability, and Distributions — Prequiz

Open-note. Show your mathematical work. A calculation and a clearly stated final answer are enough unless a sentence is specifically requested.

Name:

Date:

Useful facts

Relative frequency = $\frac{\text{count}}{\text{sample size}}$. A probability distribution has nonnegative entries that add to 1. Also, $P(\text{not } A) = 1 - P(A)$.

1. A college wants to estimate how many hours its students work for pay each week. An instructor surveys 150 students who are enrolled in classes meeting before 9 a.m.
 - (a) What is the population the college wants to understand?
 - (b) What is the sample actually observed?
 - (c) What part of the student population is not represented by this selection rule, and why might that matter when using the sample to describe the whole college?

2. In a transportation survey of 24 employees, 12 drive, 6 take the bus, 4 bike, and 2 walk.
 - (a) Find the relative frequency of taking the bus.
 - (b) Find the relative frequency of *not* driving.
 - (c) Which display is more appropriate for these data: a bar chart or a histogram? Explain with one short phrase.

3. A community center records the ages of volunteers and groups them into the intervals
 $[20, 25)$, $[25, 30)$, $[30, 35)$, $[35, 40)$, $[40, 45)$.

Which display is more appropriate for these grouped numerical data: a bar chart or a histogram? State the feature of the data that determines your choice.

4. A discrete probability distribution has four possible outcomes with probabilities

$$0.18, \quad 0.27, \quad 0.31, \quad p.$$

Find p . Then find the probability of *not* getting the outcome whose probability is 0.18.

5. Two fair six-sided dice are rolled. There are 36 equally likely ordered outcomes, and 6 of them have a sum of 7. Find $P(\text{sum} = 7)$ as a fraction and as a decimal.

Module 2: Frequency, Probability, and Distributions — Matching Quiz

These problems use the same skills as the prequiz, with new values and contexts. Show enough work to make your reasoning visible.

Name:

Date:

1. A county wants to estimate support for a new park levy. A polling organization randomly selects 500 registered county voters. Identify the population and the sample. What feature of the selection plan is intended to give a broad cross-section of the registered-voter population?
2. In a sample of 60 households, 18 report using public transportation at least once each week. Find the relative frequency of households in the sample that use public transportation weekly.
3. A researcher groups commute times into 10-minute intervals. Should the frequencies be displayed with a bar chart for categories or with a histogram? Explain briefly.
4. A probability distribution has probabilities 0.22, 0.35, 0.19, p . Find p , and then find the complement of the event with probability 0.35.
5. A spinner has 8 equal sections, of which 3 are blue. Find the probability of landing on blue and the probability of not landing on blue.